

DESIGN VERIFICATION REPORT

Designer/ Manufacturer : Special and Safety Work (SSW), Riyadh, Saudi Arabia.
DNV Project No : PP119372
Equipment Description : Anti-blast sliding door
Codes and Standards : AISC (9th Edition)

This is to confirm that the structural design of the Anti-blast sliding door (as per drawings listed in Table-1:Design Examination Document) has been reviewed and found to be in compliance with the applicable codes and standards mentioned in this Design Verification Report.

Scope of Verification:

The verification of the design of Door is limited to review of structural analysis submitted in compliance with AISC (9th Edition) as applicable with regards to local in-place strength against uniformly distributed static equivalent peak over pressure of 2 barg.

No other codes / standards have been referred to and it is concluded that client would satisfy themselves that requirements of other relevant codes / standards / national regulations e.g. those mentioned as reference in design calculations submitted etc., are complied with suitably.

DNV review is limited to the door itself. This review does not cover the design of concrete supports around the door.

Table-1: Design Examination Document:

S. No.	Drawing No.	Revision	Drawing Title	Status
1.	MOD/SSW/BLA/SH-01	01	Anti Blast Sliding Door- Exterior Elevation and Section Plan	Reviewed
2.	MOD/SSW/BLA/SH-02	01	Anti Blast Sliding Door- Exterior Elevation and Section Plan	Reviewed
3.	MOD/SSW/BLA/SH-03	01	Anti Blast Sliding Door- Details - Sheet 1/2	Reviewed
4.	MOD/SSW/BLA/SH-04	01	Anti Blast Sliding Door- Details - Sheet 2/2	Reviewed

Verification Limitations:

Design Parameters:

Parameter	Description
Blast Pressure	2 bar
Overall Door Size	6880mm width and 4100mm height
Rebound Pressure adopted	20% of blast load

Other design data are considered as per drawing listed in Table-1.



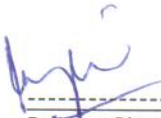
Limitations and Notes:

- (1) The design appraisal is limited to the static in place analysis of the steel door frame and excludes any aspects towards adequacy of anchoring arrangement, inspection and testing. The scope of verification is listed as above.
- (2) Door face steel sheet must have yield strength not less than 235 MPa as used in design calculation. All material used for door and the hardware components like door frame anchors, tower bolts, hinges, etc. including sealants and panic devices along with fabrication and installation arrangements shall meet the blast rating specifications and structural quality as per the client requirements.
- (3) Present appraisal has been carried out with the blast load indicated in the submitted calculations. Other loads/ load factors like brittle fracture, etc are not considered in this review.
- (4) The deflection calculations for the door has not been provided for review. In this regards we have noted designers response that the deflection is not required to be assessed for the door under consideration. We conclude that designer has satisfied themselves that deflection would not be relevant criterion for adequacy assessment.

Validity Period:

No validity period is stated on the design verification report as it only confirms the conformity with specified regulations / standards at the time of issue. Change in regulations or change in design may have influence on the validity.

Signed for **Det Norske Veritas AS, Abu Dhabi Branch**



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OPMAE 646



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Date: 20th January 2015
Place: Dubai, UAE